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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: The Mole Concept

Useful values: Avogadro constant L (N_A) = 6.02 × 10^{23} mol^{-1}; molar gas volume = 22.4 dm^3 mol^{-1} at s.t.p. (0 °C, 1 atm) and 24.0 dm^3 mol^{-1} at r.t.p. (25 °C, 1 atm); R = 8.31 J K^{-1} mol^{-1}. Relative atomic masses: H 1, C 12, N 14, O 16, Na 23, Mg 24, Al 27, S 32, Cl 35.5, K 39, Ca 40, Mn 55, Fe 56, Cu 63.5, Zn 65, Ag 108, I 127, Ba 137.

Lesson	Title (MINESEC progression)
1	The mole, Avogadro constant, relative atomic and molecular masses, molar mass
2	Calculations with moles, molar mass and the Avogadro constant
3	Empirical and molecular formulae
4	Concentration of solutions
5	Molar interpretation of chemical equations
6	Limiting reagents and percentage yield
7	Titrations: terms and equipment
8	Strong acid–strong base titrations
9	Weak acid–strong base and weak acid–weak base titrations
10	Oxidation numbers; oxidation and reduction half-equations
11	Writing and balancing redox equations
12	Redox titrations: permanganometry and dichrometry
13	Iodometry and iodimetry
14	Precipitation and complexometric titrations
15	Back titration
16	Gas laws, ideal gas equation, molar volumes
17	Determining the relative molecular mass of a volatile liquid

Lesson 1: The mole, Avogadro constant, relative atomic and molecular masses, molar mass

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define the mole, the Avogadro constant, A_r, M_r and molar mass
- ✔ calculate relative molecular masses

Lesson notes

KEY POINTS

Relative atomic mass A_r: the weighted average mass of an atom of an element compared with 1/12 of the mass of an atom of carbon-12

Relative molecular mass M_r: the mass of a molecule compared with 1/12 of the mass of a carbon-12 atom (the sum of the A_r values). For ionic compounds the term **relative formula mass** is used.

The mole: the amount of substance containing as many particles (atoms, molecules, ions, electrons) as there are atoms in exactly 12 g of carbon-12; that number is the **Avogadro constant**, 6.02 × 10^{23} mol^{-1}

Molar mass M: the mass of one mole of a substance (g mol^{-1}), numerically equal to A_r or M_r

Worked examples

Example: Find the relative molecular mass of (a) H_2SO_4 (b) $CuSO_4.5H_2O$.

(a) 2(1) + 32 + 4(16) = 98

(b) 63.5 + 32 + 64 + 5(18) = 249.5

Practice Questions: Lesson 1

1. Calculate M_r of (a) H_2SO_4 (b) $CuSO_4 \cdot 5H_2O$ (H = 1, O = 16, S = 32, Cu = 63.5).
2. Define the relative atomic mass of an element.

Answers: *1. (a) 98 (b) 249.5 2. the average mass of one atom of the element relative to 1/12 of the mass of one atom of carbon-12*

Lesson 2: Calculations with moles, molar mass and the Avogadro constant

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ convert between mass, moles and number of particles

Lesson notes

KEY POINTS

n = m/M; number of particles = n × L

Worked examples

Example: How many moles and molecules are in 11 g of CO_2?

n = 11/44 = 0.25 mol; molecules = $0.25 \times 6.02 \times 10^{23} = 1.51 \times 10^{23}$

Example: Find the mass of 0.15 mol of calcium carbonate and the number of oxygen atoms in it.

M($CaCO_3$) = 100 g mol^{-1}; m = 0.15 × 100 = 15 g

O atoms = $3 \times 0.15 \times 6.02 \times 10^{23} = 2.71 \times 10^{23}$

Practice Questions: Lesson 2

1. How many moles are in 11 g of CO_2?
2. How many molecules are in 9.0 g of water? (L = 6.02×10^{23} mol^{-1})
3. Find the mass of 0.25 mol of NaOH.

Answers: *1. 0.25 mol 2. 3.01×10^{23} 3. 10 g*

Lesson 3: Empirical and molecular formulae

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ calculate empirical and molecular formulae

Lesson notes

KEY POINTS

Empirical formula: the simplest whole-number ratio of atoms of each element in a compound.
Molecular formula: the actual number of atoms of each element in one molecule (a whole-number multiple of the empirical formula).

Method: mass (or %) of each element → ÷ A_r (moles) → ÷ smallest value → whole-number ratio.

Worked examples

Example: A compound contains 85.7% carbon and 14.3% hydrogen. Its M_r is 56. Find its empirical and molecular formulae.

C: 85.7/12 = 7.14; H: 14.3/1 = 14.3; ratio 1 : 2: empirical formula CH_2 (mass 14)

56/14 = 4: molecular formula C_4H_8

Example: 2.43 g of magnesium burns to give 4.03 g of oxide. Find the formula of the oxide.

Mg: 2.43/24 = 0.101 mol; O: (4.03 − 2.43)/16 = 0.100 mol; ratio 1 : 1: MgO

Practice Questions: Lesson 3

1. A compound contains 40.0% C, 6.7% H and 53.3% O. Its M_r is 180. Find its empirical and molecular formulae.

Answers: *1. CH_2O; $C_6H_{12}O_6$*

Lesson 4: Concentration of solutions

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ calculate concentrations in mol dm^{-3} and g dm^{-3}
- ✔ perform dilution calculations

Lesson notes

KEY POINTS

Concentration (mol dm^{-3}) = moles of solute ÷ volume of solution (dm^3); concentration (g dm^{-3}) = mol dm^{-3} × M

Dilution: $c_1V_1 = c_2V_2$ (moles of solute unchanged); 1 dm^3 = 1000 cm^3

Worked examples

Example: 4.0 g of NaOH is dissolved to make 250 cm^3 of solution. Find its concentration in mol dm^{-3} and g dm^{-3}.

n = 4.0/40 = 0.10 mol; c = 0.10/0.250 = 0.40 mol dm^{-3}; = 0.40 × 40 = 16 g dm^{-3}

Example: What volume of 2.0 mol dm^{-3} HCl is needed to make 500 cm^3 of 0.10 mol dm^{-3} HCl?

V_1 = 0.10 × 500/2.0 = 25 cm^3

Practice Questions: Lesson 4

1. 4.0 g of NaOH is dissolved to make 250 cm^3 of solution. Find the concentration in mol dm^{-3} and g dm^{-3}.
2. What volume of water must be added to 50 cm^3 of 2.0 mol dm^{-3} HCl to make 0.50 mol dm^{-3}?

Answers: *1. 0.40 mol dm^{-3}; 16 g dm^{-3} 2. 150 cm^3*

Lesson 5: Molar interpretation of chemical equations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ use balanced equations to calculate reacting quantities

Lesson notes

A balanced equation gives the mole ratio in which substances react and are produced.

Worked examples

Example: What mass of iron(III) oxide is needed to make 112 g of iron? $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

n(Fe) = 112/56 = 2.0 mol; n(Fe_2O_3) = 1.0 mol; mass = 1.0 × 160 = 160 g

Example: What volume of CO_2 at r.t.p. is formed when 25 g of $CaCO_3$ reacts with excess acid?

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$; n($CaCO_3$) = 0.25 mol = n($CO_2$)

V = 0.25 × 24.0 = 6.0 dm^3

Practice Questions: Lesson 5

1. What mass of CaO is formed when 25 g of $CaCO_3$ is heated? ($CaCO_3 \rightarrow CaO + CO_2$)
2. What volume of CO_2 at r.t.p. is produced? (24 dm^3 mol^{-1})

Answers: *1. 14 g 2. 6.0 dm^3*

Lesson 6: Limiting reagents and percentage yield

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ identify the limiting reagent
- ✔ calculate percentage yield

Lesson notes

KEY POINTS

The **limiting reagent** is completely used up and determines the amount of product; other reactants are in excess.

Percentage yield = (actual yield ÷ theoretical yield) × 100%. Atom economy = (M_r of desired product ÷ total M_r of all products) × 100%

Worked examples

Example: 5.0 g of hydrogen reacts with 32 g of oxygen: $2H_2 + O_2 \rightarrow 2H_2O$. Find the limiting reagent and the mass of water formed.

$n(H_2) = 2.5$ mol needs 1.25 mol O_2; only 1.0 mol O_2 is present: O_2 is limiting

$n(H_2O) = 2 \times 1.0 = 2.0$ mol = 36 g

Example: Heating 20 g of $CaCO_3$ gives 9.8 g of CaO. Find the percentage yield.

Theoretical: 0.20 mol CaO = 11.2 g; yield = 9.8/11.2 × 100 = 87.5%

Practice Questions: Lesson 6

1. 2.4 g of Mg reacts with 0.15 mol of HCl. Which is limiting? ($Mg + 2HCl \rightarrow MgCl_2 + H_2$)
2. 10 g of $CaCO_3$ gives 4.8 g of CaO. Find the percentage yield.

Answers: *1. HCl (0.10 mol Mg needs 0.20 mol HCl) 2. 85.7%*

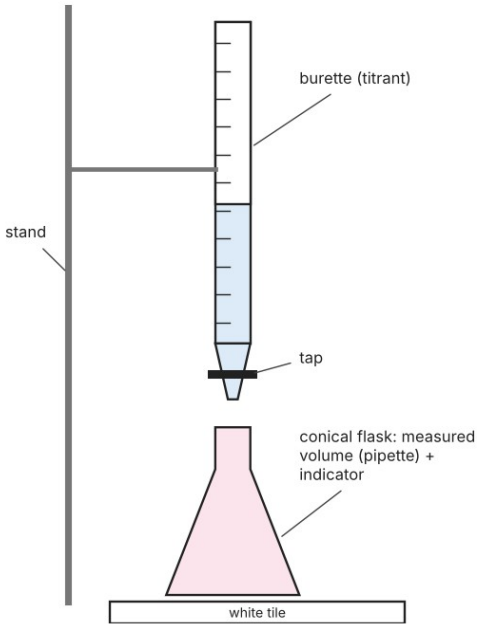
Lesson 7: Titrations: terms and equipment

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ define standard solution and primary standard
- ✔ name titration apparatus and describe how it is used

Lesson notes



- **Titration:** the gradual addition of one solution (from a burette) to a known volume of another (pipetted into a conical flask) until the reaction is just complete, shown by an **indicator** (end point).
- **Standard solution:** a solution of accurately known concentration. A **primary standard** is a substance used to prepare one directly by weighing: pure, stable in air (not hygroscopic, not reacting with CO_2), high molar mass, soluble (e.g. anhydrous Na_2CO_3, potassium hydrogen phthalate, oxalic acid). NaOH is NOT a primary standard (absorbs water and CO_2).
- **Equipment:** pipette (25.0 cm^3), burette (50.00 cm^3, read to 0.05 cm^3), volumetric flask (250 cm^3), conical flask, white tile, wash bottle, funnel.
- **Good practice:** rinse the pipette and burette with the solutions they will hold, the flask with distilled water; read the bottom of the meniscus at eye level; do a rough titration then repeat until two concordant titres (within 0.10 cm^3); average them.

Practice Questions: Lesson 7

1. State three properties of a primary standard.
2. Why is the burette rinsed with the solution it will contain?

Answers: *1. high purity; stable in air (not hygroscopic); high molar mass; soluble 2. so that water does not dilute the solution*

Lesson 8: Strong acid–strong base titrations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ carry out calculations from strong acid–strong base titrations

Worked examples

Example: 25.0 cm^3 of NaOH solution needs 20.0 cm^3 of 0.150 mol dm^{-3} HCl for neutralisation. Find the concentration of the NaOH.

$NaOH + HCl \rightarrow NaCl + H_2O$

$n(HCl) = 0.0200 \times 0.150 = 3.00 \times 10^{-3}$ mol = n(NaOH)

$[NaOH] = 3.00 \times 10^{-3} / 0.0250 = 0.120$ mol dm^{-3}

Example: 25.0 cm^3 of 0.100 mol dm^{-3} H_2SO_4 is neutralised by 31.2 cm^3 of KOH. Find [KOH].

$n(H_2SO_4) = 2.50 \times 10^{-3}$ mol; $n(KOH) = 2 \times 2.50 \times 10^{-3} = 5.00 \times 10^{-3}$ mol

$[KOH] = 5.00 \times 10^{-3} / 0.0312 = 0.160$ mol dm^{-3}

Practice Questions: Lesson 8

1. 25.0 cm^3 of NaOH needed 20.0 cm^3 of 0.10 mol dm^{-3} HCl. Find [NaOH].
2. 25.0 cm^3 of H_2SO_4 needed 30.0 cm^3 of 0.20 mol dm^{-3} NaOH. Find $[H_2SO_4]$.

Answers: *1. 0.080 mol dm^{-3} 2. 0.12 mol dm^{-3}*

Lesson 9: Weak acid–strong base and weak acid–weak base titrations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ choose suitable indicators for different titrations
- ✔ calculate concentrations of weak acids

Lesson notes

- Choice of indicator depends on the pH at the equivalence point: strong acid–strong base (pH 7): methyl orange or phenolphthalein; weak acid–strong base (pH about 9): **phenolphthalein**; strong acid–weak base (pH about 5): **methyl orange**; weak acid–weak base: no sharp end point (use a pH meter or conductivity).

Worked examples

Example: 10.0 cm^3 of vinegar is diluted to 100 cm^3. 25.0 cm^3 of the diluted solution needs 21.0 cm^3 of 0.100 mol dm^{-3} NaOH (phenolphthalein). Find the concentration of ethanoic acid in the vinegar in g dm^{-3}.

$n(NaOH) = 2.10 \times 10^{-3}$ mol = $n(CH_3COOH)$ in 25.0 cm^3

In 100 cm^3: 8.40×10^{-3} mol (all from 10.0 cm^3 of vinegar)

$[CH_3COOH] = 8.40 \times 10^{-3} / 0.0100 = 0.840$ mol dm^{-3} $= 0.840 \times 60 = 50.4$ g dm^{-3}

Practice Questions: Lesson 9

1. Which indicator is suitable for ethanoic acid–NaOH titration? Why?
2. 25.0 cm^3 of ethanoic acid needed 22.5 cm^3 of 0.10 mol dm^{-3} NaOH. Find the concentration of the acid.

Answers: *1. phenolphthalein: the end point is alkaline (about pH 9) 2. 0.090 mol* dm^{-3}

Lesson 10: Oxidation numbers; oxidation and reduction half-equations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ assign oxidation numbers
- ✔ write oxidation and reduction half-equations

Lesson notes

KEY POINTS

Oxidation = loss of electrons / increase in oxidation number. **Reduction** = gain of electrons / decrease in oxidation number ("OIL RIG")

Rules for oxidation numbers:

- Uncombined elements: 0 (Na, O_2, Cl_2).
- Simple ions: the charge ($Na^+ = +1$, $Cl^- = -1$).
- In compounds: O is −2 (except peroxides −1, OF_2 +2); H is +1 (except metal hydrides −1); group 1 metals +1, group 2 +2; F is −1.
- The sum of oxidation numbers in a compound is 0; in an ion, it equals the charge.

Worked examples

Example: Find the oxidation number of Mn in $KMnO_4$, Cr in $K_2Cr_2O_7$ and S in $S_2O_3^{2-}$.

Mn: $+1 + x + 4(-2) = 0$, $x = +7$

Cr: $2(+1) + 2x + 7(-2) = 0$, $x = +6$

S: $2x + 3(-2) = -2$, $x = +2$

Example: Write half-equations for: (a) $Fe^{2+} \rightarrow Fe^{3+}$ (b) $Cl_2 \rightarrow Cl^-$.

(a) $Fe^{2+} \rightarrow Fe^{3+} + e^-$ (oxidation)

(b) $Cl_2 + 2e^- \rightarrow 2Cl^-$ (reduction)

Practice Questions: Lesson 10

1. Find the oxidation number of Mn in $KMnO_4$ and of Cr in $K_2Cr_2O_7$.
2. Write the half-equation for $Fe^{2+} \rightarrow Fe^{3+}$.

Answers: *1. +7; +6 2. $Fe^{2+} \rightarrow Fe^{3+} + e^-$*

Lesson 11: Writing and balancing redox equations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ balance redox equations using half-equations

Lesson notes

Method (half-equations in acid): balance atoms other than O and H; add H_2O to balance O; add H^+ to balance H; add electrons to balance charge; multiply the half-equations so the electrons cancel; add.

Worked examples

Example: Balance the reduction of MnO_4^- to Mn^{2+} in acid and combine with $Fe^{2+} \rightarrow Fe^{3+}$.

$MnO_4^- \rightarrow Mn^{2+} + 4H_2O$ (balance O); $MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$ (balance H)

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ (charge: +7 on the left → +2)

× 1 and $Fe^{2+} \rightarrow Fe^{3+} + e^-$ × 5:

$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

Example: Balance $Cr_2O_7^{2-} \rightarrow Cr^{3+}$ in acid.

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

Practice Questions: Lesson 11

1. Balance: $MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2$ (acid solution).

Answers: *1. $2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$*

Lesson 12: Redox titrations: permanganometry and dichrometry

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ carry out calculations on permanganate and dichromate titrations

Lesson notes

- **Potassium manganate(VII) ($KMnO_4$):** purple; acts as its own indicator (end point: first permanent pale pink). Acidified with dilute H_2SO_4 (not HCl, which would be oxidised). Reacts with Fe^{2+} (1 : 5), ethanedioate $C_2O_4^{2-}$ (2 : 5, warm to 60 °C), H_2O_2 (2 : 5).
- **Potassium dichromate(VI) ($K_2Cr_2O_7$):** orange → green Cr^{3+}; a primary standard; needs an indicator (e.g. diphenylamine sulphonate). Reacts with Fe^{2+} in the ratio 1 : 6.

Worked examples

Example: 25.0 cm^3 of an iron(II) solution needs 22.0 cm^3 of 0.0200 mol dm^{-3} $KMnO_4$. Find $[Fe^{2+}]$.

$n(MnO_4^-) = 0.0220 \times 0.0200 = 4.40 \times 10^{-4}$ mol; $n(Fe^{2+}) = 5 \times 4.40 \times 10^{-4} = 2.20 \times 10^{-3}$ mol

$[Fe^{2+}] = 2.20 \times 10^{-3} / 0.0250 = 0.0880$ mol dm^{-3}

Example: 1.50 g of iron ore was dissolved and all the iron converted to Fe^{2+}, which needed 24.0 cm^3 of 0.0200 mol dm^{-3} $K_2Cr_2O_7$. Find the percentage of iron in the ore.

$n(Cr_2O_7^{2-}) = 4.80 \times 10^{-4}$ mol; $n(Fe^{2+}) = 6 \times 4.80 \times 10^{-4} = 2.88 \times 10^{-3}$ mol

mass Fe = $2.88 \times 10^{-3} \times 56 = 0.161$ g; % = $0.161 / 1.50 \times 100 = 10.8\%$

Practice Questions: Lesson 12

1. 25.0 cm^3 of Fe^{2+} solution needed 20.0 cm^3 of 0.020 mol dm^{-3} $KMnO_4$. Find $[Fe^{2+}]$.
2. Why is no indicator needed in $KMnO_4$ titrations?

Answers: *1. 0.080 mol dm^{-3} 2. MnO_4^- is purple; the first excess drop gives a permanent pink colour*

Lesson 13: Iodometry and iodimetry

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish iodometry from iodimetry
- ✔ carry out iodine–thiosulphate calculations

Lesson notes

- **Iodometry:** an oxidising agent liberates iodine from excess KI; the iodine is titrated with standard sodium thiosulphate: $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$. Starch indicator is added near the end point (blue-black → colourless).
- **Iodimetry:** direct titration of a reducing agent with a standard iodine solution.

Worked examples

Example: 25.0 cm^3 of a solution of KIO_3 is added to excess KI and acid: $IO_3^- + 5I^- + 6H^+ \rightarrow 3I_2 + 3H_2O$. The iodine needs 24.0 cm^3 of 0.100 mol dm^{-3} $Na_2S_2O_3$. Find $[IO_3^-]$.

$n(S_2O_3^{2-}) = 2.40 \times 10^{-3}$ mol; $n(I_2) = 1.20 \times 10^{-3}$ mol; $n(IO_3^-) = 1.20 \times 10^{-3}/3 = 4.00 \times 10^{-4}$ mol

$[IO_3^-] = 4.00 \times 10^{-4}/0.0250 = 0.0160$ mol dm^{-3}

Practice Questions: Lesson 13

1. Iodine liberated from excess KI needed 24.0 cm^3 of 0.10 mol dm^{-3} $Na_2S_2O_3$. How many moles of I_2 were formed?
2. Why is starch added near the end point?

Answers: *1. 1.20×10^{-3} mol 2. starch–iodine complex forms early and may not release iodine readily; adding it late gives a sharp change*

Lesson 14: Precipitation and complexometric titrations

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ carry out calculations on precipitation and EDTA titrations

Lesson notes

- **Precipitation titration (argentometry, Mohr's method):** chloride ions are titrated with standard silver nitrate: $Ag^+ + Cl^- \rightarrow AgCl(s)$ (white). Potassium chromate indicator gives a red-brown precipitate of Ag_2CrO_4 at the end point.
- **Complexometric titration:** metal ions (Ca^{2+}, Mg^{2+}: water hardness) are titrated with EDTA, which forms 1 : 1 complexes; indicator Eriochrome Black T (wine red → blue).

Worked examples

Example: 25.0 cm^3 of a salt solution needs 18.5 cm^3 of 0.100 mol dm^{-3} $AgNO_3$. Find the concentration of NaCl in g dm^{-3}.

$n(Ag^+) = 1.85 \times 10^{-3}$ mol = $n(Cl^-)$; [NaCl] = $1.85 \times 10^{-3}/0.0250 = 0.0740$ mol $dm^{-3} = 0.0740 \times 58.5 = 4.33$ g dm^{-3}

Example: 50.0 cm^3 of tap water needs 12.0 cm^3 of 0.0100 mol dm^{-3} EDTA. Find the hardness as mg of $CaCO_3$ per dm^3.

$n(EDTA) = 1.20 \times 10^{-4}$ mol = $n(Ca^{2+})$; per dm^3: 2.40×10^{-3} mol = 0.240 g = 240 mg $CaCO_3$ dm^{-3}

Practice Questions: Lesson 14

1. 25.0 cm^3 of NaCl needed 18.0 cm^3 of 0.050 mol dm^{-3} $AgNO_3$. Find [NaCl].
2. 50.0 cm^3 of hard water needed 12.0 cm^3 of 0.010 mol dm^{-3} EDTA. Find $[Ca^{2+}]$.

Answers: *1. 0.036 mol dm^{-3} 2. 2.4×10^{-3} mol dm^{-3}*

END OF FREE SAMPLE

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